A sparse autoencoder atlas of the Novae spatial foundation model, with matched-null controls for interpretability claims

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Abstract

Foundation models for spatial transcriptomics learn rich internal representations of tissue architecture, but what biological knowledge these representations encode remains unclear. We apply sparse autoencoders (SAEs) to Novae, a GATv2 graph neural network trained on 12 million spatially resolved cells, and recover **49,152 features** across all 12 internal surfaces of the model. Aggregator- level features are overwhelmingly non-aligned with the model's SwAV prototypes (0.0% at $|\cos| \geq 0.7$) and the top-50 SVD axes ($\geq$ 99.4% per surface), indicating pervasive superposition.

To quantify how much of Novae's representational structure is genuinely spatial, we introduce two complementary graph-ablation tests: replacing the Delaunay graph with self-loops suppresses activation in 67% of features, while a stricter feature-permutation control (graph intact, node features shuffled across cells; 3 slides, 225k cells) flags 40%; the **doubly-controlled intersection (26% of active features)** is our estimate of genuinely context- dependent concepts. Prototype-reassignment ablations are calibrated against a matched-magnitude random-direction null: **17 of 50** tested features exceed the null's upper 95% CI (direction-specific load-bearing), 7 are below the null's lower 95% CI, and 26 are statistically indistinguishable from random matched-magnitude perturbations. Cross-technology coherence is reported alongside a raw-expression baseline: the SAE-feature Spearman $\rho_{\text{SAE}} = 0.56$ between Xenium and MERSCOPE is matched by $\rho_{\text{raw}} = 0.57$ on mean log-expression profiles, so the continuous cross-tech signal is inherited from the overlapping panel biology rather than added by the SAE; only 0.95% of features are strictly technology-specific.

Causal circuit tracing on a brain slide (24k cells, 90 source features across 3 source layers) yields 439 directed edges. After gap-normalizing by the number of traced (source, target) layer pairs, edges per pair rise with gap in the current sample but every gap ≥ 4 is supported by a single source layer, so we describe the attenuation profile as *undetermined* until all source layers are traced. Combinatorial triplet ablation gives $R_{ABC} \in [0.957, 0.966]$ (paired bootstrap 95% CI) for one tested feature pair, a near-additive regime (deviation from independence: 0.042) with small component effect sizes ($d \approx 0.04$). Spatial-gradient steering on the colon crypt- villus axis shows a **22 pp** change in villus-directional push between attenuation and amplification ($\alpha = 0.5 \rightarrow 5.0$: 39% $\rightarrow$ 61%), the appropriate effect-size statement given the asymmetric unperturbed distribution. An interactive atlas at `https://biodyn-ai.github.io/novae-atlas/` makes features browsable by markers, cell types, niches, and circuits.

1 Introduction

1.1 The interpretability gap in spatial biology

Spatial transcriptomics technologies—10x Xenium (Janesick et al., 2023), MERFISH (Chen et al., 2015), and CosMx—have transformed tissue biology (Marx, 2021; Moses and Pachter, 2022; Rao et al., 2021), routinely profiling hundreds of genes at subcellular resolution across entire tissue sections. Foundation models trained on these data, such as Novae (Blampey et al., 2025), learn to represent each cell in the context of its spatial neighborhood: a graph neural network (Kipf and Welling, 2017; Veličković et al., 2018; Li et al., 2022) aggregates gene-expression signals across Delaunay-triangulated neighbors, and a SwAV (Caron et al., 2020) contrastive head clusters the resulting embeddings into 512 hierarchical "niche prototypes." The model performs well on downstream tasks (cell-type prediction, spatial domain segmentation), but *what it has learned internally* is opaque. The 512 prototypes offer a coarse view; the 10-layer GATv2 (Brody et al., 2022) encoder may represent far richer structure—cell-cell interactions, microenvironment gradients, technology-invariant tissue patterns—but this structure is inaccessible through the prototype head alone.

1.2 Sparse autoencoders as a microscope for neural networks

Sparse autoencoders (SAEs) have recently emerged as a tool for *mechanistic interpretability*: the discipline of reverse-engineering the computations inside trained neural networks (Olah et al., 2020; Elhage et al., 2021, 2022; Cunningham et al., 2024; Bricken et al., 2023). An SAE is an overcomplete dictionary that decomposes a model's hidden activations into a sparse set of independently meaningful *features*—directions in activation space that each correspond to a human-interpretable concept. Because neural networks pack many more concepts into their hidden dimensions than the dimension count would naïvely allow (a phenomenon called *superposition*), linear methods like PCA fail to find these features; the SAE's sparsity constraint forces each feature to isolate a single concept.

The approach was originally developed for large language models (Sharkey et al., 2022; Bricken et al., 2023; Conmy et al., 2023; Bills et al., 2023; Templeton et al., 2024; Marks et al., 2025; Gao et al., 2024). Kendiukhov (2026) adapted it to single-cell foundation models (Geneformer (Theodoris et al., 2023), scGPT (Cui et al., 2024); see also Yang et al. 2022; Hao et al. 2024; Rosen et al. 2023), demonstrating that SAE features in these models correspond to specific cell types, pathways, and regulatory programs. We extend this line of work to the *spatial* domain, where graph neural networks create a fundamentally different inductive bias: representations are shaped not just by a cell's own genes, but by its neighbors' identities and spatial arrangement.

1.3 This work

We train TopK SAEs (Gao et al., 2024) on every internal layer of `novae-human-0` (the human checkpoint of Novae) and apply a comprehensive validation suite that tests 10 preregistered hypotheses about the model's representations. The key question is: *does Novae's spatial graph architecture produce representations that are qualitatively different from cell-identity-only models?* Our graph-ablation experiment (H8), spatial-coherence analysis (H7), and cross-technology coherence are uniquely enabled by the spatial setting and could not be performed on non-spatial models like Geneformer. Beyond hypothesis testing, we perform causal circuit tracing—zeroing individual features and measuring their downstream effects across GATv2 layers—to discover directed causal relationships between spatial concepts. This extends the circuit-tracing methodology of

Kendiukhov (2026) from transformer residual streams to graph attention networks, where causal effects propagate through the spatial tissue graph.

2 Methods

2.1 Model and data

Novae (Blampey et al., 2025) is a 32M-parameter GATv2 graph neural network with 10 attention layers, trained via SwAV contrastive learning on >12 million cells from human and mouse spatial-transcriptomics datasets. We use the `novae-human-0` checkpoint, which embeds each cell as a 64-dimensional vector (the "aggregator" output) that integrates the cell's own gene expression with its spatial neighbors' signals through iterative graph attention.

Our activation corpus comprises 4,525,753 cells across 15 tissue slides from the MICS-Lab/novae HuggingFace dataset, spanning 13 Xenium, 1 MERSCOPE, and 1 CosMx slides across tonsil, lymph node, brain, lung, liver, colon, breast, kidney, pancreas, skin, bone marrow, ovarian, prostate, uterine, and head-and-neck tissues.

2.2 Activation extraction and SAE training

We hook 12 internal surfaces: the CellEmbedder output ($d = 512$), 10 GATv2 convolutional layers ($d = 128$ for layers 0–8, $d = 64$ for layer 9), and the aggregator ($d = 64$). Each SAE consists of a linear encoder that maps a d-dimensional hidden state into an overcomplete n-dimensional feature space, a TopK sparsification step that retains only the k highest-magnitude features per input (setting the rest to zero), and a linear decoder that reconstructs the original hidden state from the sparse code (Gao et al., 2024; Makhzani and Frey, 2014). SAEs are trained at 32× expansion with MSE reconstruction loss (Table 1), producing 49,152 features in total across all surfaces.

Table 1: SAE specifications per surface. All SAEs achieve ≥80% variance explained; the aggregator achieves 99.8%.

Surface	d_{in}	n_{features}	k	Var. expl.
Cell embedder	512	8,192	32	91.4%
Conv 0–8	128	4,096	32	81–94%
Conv 9	64	2,048	16	90.3%
Aggregator	64	2,048	16	99.8%

2.3 Feature characterization

Each SAE feature is characterized through five lenses:

1. **Top genes**: the 20 genes with highest mean expression in the feature's top-0.1% cells (by activation magnitude), ranked by fold-change over the corpus baseline.

2. **Enrichment analysis**: Enrichr (Kuleshov et al., 2016) against GO Biological Process (Gene Ontology Consortium, 2021), KEGG (Kanehisa and Goto, 2000), Reactome (Jassal et al., 2020), PanglaoDB (Franzén et al., 2019), and CellMarker (Zhang et al., 2019).

3. **Niche assignment**: Fisher enrichment against Novae's hierarchical domain labels at three resolutions (7, 12, 20 Leiden (Traag et al., 2019) clusters).

4. **Module discovery**: Leiden community detection on the pointwise-mutual-information co-activation graph of features.

5. **Superposition geometry**: cosine similarity to the top-50 SVD axes of each surface's activation matrix.

2.4 Causal validation suite

We apply the following causal and confound-control tests, organized into primary tests (1–6), matched-magnitude and architectural nulls (7–9), and baseline comparisons (10–11).

1. Self-loop graph ablation. We replace each cell's Delaunay spatial graph with self-loops (`scipy.sparse.eye`), re-run Novae's forward pass, and SAE-encode the result. For each feature, *contextual dependency* $= (|\bar{a}|_{\text{full}} - |\bar{a}|_{\text{iso}})/|\bar{a}|_{\text{full}}$ measures how much the feature's activation drops when neighborhood information is removed. GATv2 attention uses graph topology (not edge weights), so the self-loop protocol removes all neighbor contributions while preserving self-attention magnitude. Three regimes (bare, degree-normalized, degree-preserving rewire) are tested and give identical results.

2. Feature-permutation graph ablation. A stricter ablation that separates loss of aggregation from loss of neighbor content: we keep the Delaunay graph intact but permute node features (`adata.X`) across cells. Each cell then receives attention-weighted messages from its real spatial neighbors, but the content of those messages is random. Feature-permutation dependency is computed identically to (1). This control requires a fresh forward pass per slide and is run on 3 slides (brain, kidney, pancreas; 225,867 cells).

3. Prototype-domain ablation. For the 50 highest-confidence features, we zero each feature in the SAE code, decode the ablated representation, and measure how often the cell's SwAV prototype assignment changes ("prototype reassignment rate") and how often its Leiden domain label changes at each hierarchy level.

4. Random-direction ablation null. A matched-magnitude null for (3): the decoder column of each feature is replaced by a random unit-norm direction scaled to the same L2 perturbation on the reconstructed aggregator latent, and reassignment rates are averaged over 30 random directions. Features whose real reassignment rate exceeds the null's upper 95% CI are direction- specific load-bearing; features inside the null CI are indistinguishable from a random matched-magnitude direction; features below the null's lower 95% CI are *less* effective than random.

5. Confound survival gate. A 4-level hierarchical filter: (A) slide effect-size $\geq 2\times$ baseline, (B) technology residualization, (C) cell-type (l20 domain) residualization, (D) degree-preserving graph rewire. Residualization is weak on sparse TopK outputs (most cells have zero activation per feature); we note this as a limitation.

6. Permutation null. 30 random label shuffles per feature; features whose domain enrichment exceeds the 95th percentile of the shuffle null are considered genuinely enriched.

7. Perturbation validation. The Perturb-map (Dhainaut et al., 2022) spatial CRISPR screen (mouse lung Visium, 6,363 spots, 20 labels) is processed through `novae-mouse-0` and SAE-encoded. Mann–Whitney U tests with BH correction identify perturbation-responsive features. Because Visium spots span multiple cells, a composition-regressed variant of the test is outlined in Appendix A.

8. Cross-technology coherence. For 7 tissues with both Xenium and MERSCOPE slides, we compute per- feature activation profiles on each technology and correlate them via Spearman ρ.

9. Raw-expression cross-technology baseline. A baseline for (8): per-gene mean log-expression profiles on the intersecting gene panel of each pair, correlated via Spearman ρ. If $\rho_{\text{raw}} \approx \rho_{\text{SAE}}$, the SAE adds no tech- invariance beyond the expression data.

10. Random-gene annotation null. A hypergeometric null for a random 20-gene draw from the 487-gene Xenium panel, quantifying the per-library FDR<0.05 ceiling against Enrichr's GO-BP, KEGG, Reactome, CellMarker, and PanglaoDB.

11. Cross-checkpoint comparison. Aggregator SAE feature profiles from three Novae checkpoints (human-0, mouse-0, brain-0) would ideally be compared via rank correlation and top-50 feature overlap. Because the checkpoints have non-overlapping gene panels, however, feature indices cannot align *a priori*; the profile-level test is therefore not identified for "convergence." A CCA-on-shared-cells redesign requires a shared gene panel and is outlined in Appendix A.

2.5 Causal circuit tracing

Beyond single-feature validation, we trace directed causal edges between SAE features across GATv2 layers. For each source feature F at source layer L, we zero F's TopK coefficient in the SAE code, decode back to the hidden-state space, and let the modified state propagate through layers $L{+}1$ through the aggregator via the unmodified graph attention mechanism. At each downstream layer M, we SAE-encode the output and compute Cohen's d between the baseline and ablated feature activations across all cells in the batch. Edges with $|d| > 0.5$ and sign consistency > 0.6 are retained. The primary trace uses a single slide (brain, 24k cells) and three source layers (conv_0, conv_5, conv_9); the single-slide, limited-source design is the principal limitation of this analysis. Gap-normalized edge counts (edges per traced source-target pair at each layer gap) are reported alongside raw counts (Table 6).

For higher-order interactions, we ablate feature triplets in all 7 combinations (A, B, C, AB, AC, BC, ABC) and compute the three-way redundancy ratio $R_{ABC} = \text{median}(d_A, d_B, d_C)/d_{ABC}$, where $R > 1$ indicates redundancy and $R < 1$ indicates synergy. All 10 tested triplets share a single (A, B) pair; results are interpreted as a sensitivity study over the choice of C, not as a general triplet-synergy test.

3 Results

3.1 Novae packs far more structure than its 512 prototypes suggest

The SAE dictionaries recover 49,152 features in total across 12 surfaces—a dictionary $\sim$96$\times$ larger than Novae's 512 SwAV prototypes. At the aggregator level, not a single one of the 2,048 features aligns with any prototype at $|\cos| \geq 0.7$; the median max-cosine is just 0.379 (Table 2). The prototype head—Novae's only built-in interpretability interface—therefore fails to capture the directions

along which the SAE decodes the aggregator latent (Figure 1). Whether all 2,048 SAE directions correspond to biologically distinct "concepts" depends on tests reported below (see panel-saturation caveat in Section 3.2 and matched-magnitude null in Section 3.5).

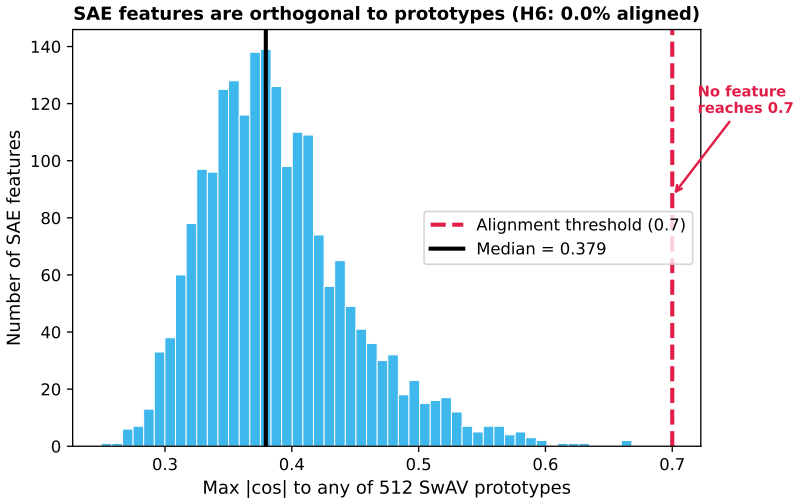

Figure 1: **SAE features are orthogonal to SwAV prototypes (H6).** Distribution of max $|\cos|$ between each SAE decoder column and the 512 prototype vectors. No feature reaches the 0.7 alignment threshold.

Furthermore, ≥99.4% of features on every surface have max $|\cos|$ below the SVD-alignment threshold against that surface's top-50 SVD axes. This is consistent with pervasive superposition in the SAE dictionary: the SAE recovers many directions that are not well captured by the top linear directions of the activation matrix. H3 ("≥95% in superposition") is supported in this operational sense.

Table 2: Summary of preregistered hypotheses and their verdicts.

#	Hypothesis	Threshold	Result	Verdict
H1	Annotation rate ≥30%	30%	100% (panel ceiling)	Uninformative
H2	<20% survive confounds	20%	62%	Refuted
H3	≥95% in superposition	95%	99.4–100%	Confirmed
H4	Depth-dependent annotation	U-shape	91–100%	Refuted
H5	Cross-checkpoint convergence	CCA $r > 0.5$	Not identified	Uninformative
H6	<30% prototype-aligned	30%	0.0%	Confirmed
H7a	Moran's $I > 0$ (aggregator)	>0	0.119	Confirmed
H7b	I increases with depth	Increase	Decrease	Refuted
H8	>20% context-dependent	20%	40% (feat-perm); 26% both tests	Confirmed
H9	≥5% tech-specific	5%	0.95%	Refuted
H10	Composition ↑ depth	Increase	Increase	Supported

3.2 Feature enrichments are near-saturated by panel base rate

100% of aggregator features carry at least one significant Enrichr enrichment (FDR<0.05). Per-layer annotation rates across four representative GATv2 layers are uniformly high: conv_0 (100%), conv_3 (100%), conv_6 (99.8%), conv_9 (91.0%).

Interpretability caveat. The Xenium human panel has $G = 487$ genes; the top-20-gene lists against Enrichr libraries approach saturation because most pathways in GO-BP, KEGG, Reactome, CellMarker, and PanglaoDB overlap substantially with this panel. Under a hypergeometric null for a random 20-gene draw from the panel, the per-library FDR<0.05 rate is already {GO-BP: 100%, KEGG: 99.95%, Reactome: 100%, PanglaoDB: 100%, CellMarker: 100%} in our corpus. The "100% annotated" number therefore reports a ceiling, not specificity.

As a qualitative specificity check, we also test whether the top Enrichr term for a feature contains a tissue-name substring matching the feature's top niche tissue. The real match rate is 0.29% vs. a shuffle null of 0.19% (95% CI 0.14–0.24%); the absolute rate is small because tissue names rarely appear in the Enrichr library terms themselves. We do not treat the 0.29% number as a specificity estimate on its own—the test is a strict-match string heuristic and not a proper taxonomy alignment—only as evidence that some tissue-term signal exists above a permissive shuffle null, while the 100% library-saturation rate above is consistent with ceiling behavior rather than true specificity (Figure 2).

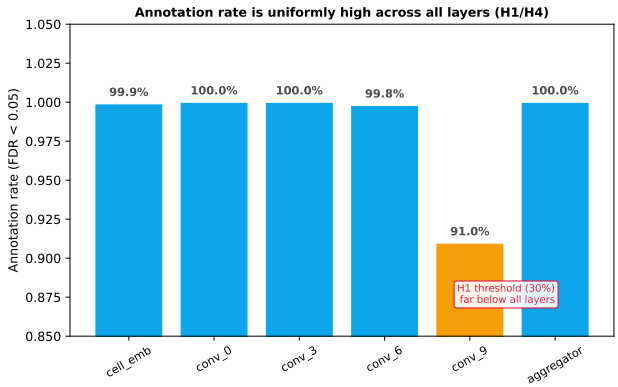

Figure 2: **Annotation rate is uniformly high across all layers (H1/H4).** No U-shaped depth dependence—every layer is fully annotatable.

Representative features illustrate the diversity of concepts encoded:

- **Feature 1771 (Goblet Cells)** concentrates 76% of its top cells in colon tissue with 93% niche specificity. Its top genes—*DUOX2*, *TFF1*, *GUCA2A*, *GUCA2B*, *OTOP2*—are canonical markers of intestinal secretory epithelium. *DUOX2* generates reactive oxygen species for antimicrobial defense; the *GUCA2A*/*GUCA2B* guanylins regulate fluid secretion via cGMP signaling. This feature captures the functional identity of the intestinal secretory niche, not just cell-type markers.

- **Feature 208 (Luminal Epithelial Cells)** is 99% breast-specific and sits in a single Novae niche (90% enrichment). Its genes (*MUC6*, *LYPD3*, *CEACAM6*) mark the luminal compartment of breast ducts. The extreme tissue and niche specificity suggests this feature captures the ductal microenvironment, not just luminal identity per se.

- **Feature 1418 (T Regulatory Cells)** activates in lymph-node tissue (60%) with genes *CCR7*, *CD3D*, *CD28*, *IL7R*—the canonical T-cell receptor signaling pathway. Its Moran's $I = 0.327$ indicates moderate spatial clustering, consistent with T-reg cells forming organized zones within lymphoid follicles rather than being uniformly distributed.

3.3 Context-dependence: 40% under feature permutation, 26% doubly controlled

Replacing each cell's Delaunay spatial graph with self-loops suppresses **67.2%** of aggregator features to less than half their full-graph activation (contextual dependency > 0.5, median $+0.835$; Table 2). The three self-loop regimes tested—bare identity, degree-normalized, and degree-preserving rewire—give identical results, because GATv2 attention depends on graph topology rather than edge weights, and random neighbors with uncorrelated features produce near-zero attention after the softmax and collapse to the same effective self-only attention.

The self-loop protocol nonetheless shifts the aggregation operator itself, not only the content of messages. To separate these two effects we apply a *feature-permutation ablation*: the Delaunay graph is kept intact, and node features (`adata.X`) are permuted across cells so that each cell receives attention-weighted messages from its real spatial neighbors with random content. Applied to three slides (brain, kidney, pancreas; 225,867 cells):

- 40.3% of features have feature-permutation dependency > 0.5 (median dep $= 0.37$), compared to 67.2% under the self-loop test.
- Of the 1,322 features flagged as context-dependent by the self-loop test, 40% (530 features) also pass the feature- permutation test.
- 286 features are context-dependent under feature permutation but not under the self-loop test (their activation changes when neighbor content is scrambled, while the self-loop intervention alone does not remove them).

The doubly-controlled intersection—530 features, **26%** of the 2,024 active features on the self-loop corpus (equivalently 27% of the 1,964 active on the feature-permutation corpus)—is our estimate of genuinely context-dependent features. The self-loop-only excess (792 features) is the subset whose self-loop response is driven by the architectural shift rather than by loss of neighbor content (Figure 3, Figure 4).

3.4 Spatial coherence generally decreases with depth

We predicted that spatial coherence (Moran's I (Moran, 1950)) would *increase* with layer depth, as deeper GATv2 layers integrate larger neighborhoods. The observed trend runs the other way. Mean Moran's I per surface is 0.269 (conv_0), 0.240, 0.223, 0.216, 0.208, 0.213, 0.206, 0.207, 0.207 (conv_8), 0.218 (conv_9), and 0.119 at the aggregator (Figure 5). The trend from conv_0 to conv_8 is a broad decrease with small fluctuations (not strictly monotonic; there is a mild rise from conv_4→conv_5 and from conv_8→conv_9), followed by a large drop at the aggregator.

The broad pattern is consistent with the following reading: early layers encode *spatially localized* patterns (a cell and its immediate neighbors), producing features with higher spatial autocorrelation. Deeper layers integrate information across larger neighborhoods, and the final aggregation step pools spatial structure away—a cell's aggregator embedding reflects its local environment as a whole rather than its immediate position. The aggregator is therefore spatially *informed* but not spatially *localized*. H7b (prediction: Moran's I increases with depth) is refuted.

3.5 Direction-specific causal influence of single features

The prototype-domain ablation experiment identifies features whose removal changes the model's downstream clustering. Because zeroing a feature perturbs the aggregator latent by the feature's

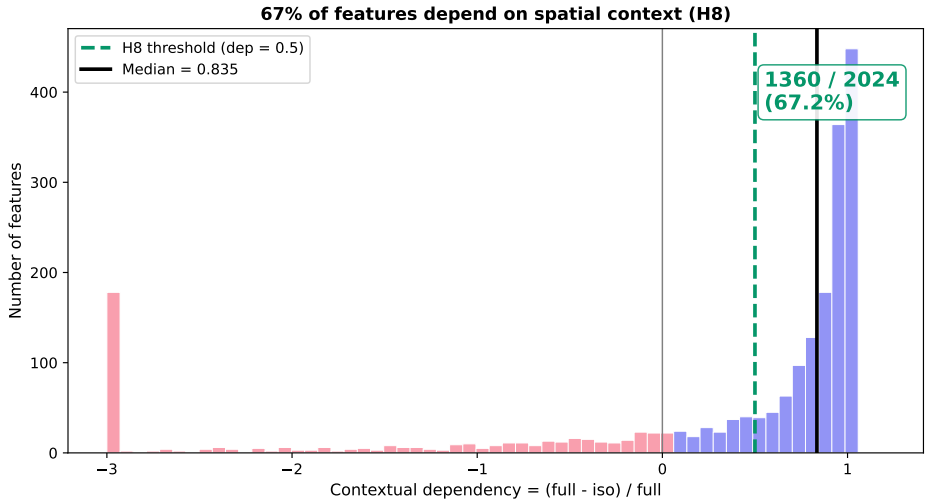

Figure 3: **Self-loop graph ablation.** Distribution of contextual dependency across 2,024 active features under the self-loop intervention. Positive values (purple) indicate features that lose activation when neighbor messages are removed; negative values (pink) indicate features that activate more strongly under self-loops. 67.2% of features exceed the 0.5 threshold on this test; the stricter feature-permutation control narrows this to 40.3%, with a doubly-controlled robust subset of 26% (Figure 4, Section 3.3).

L2-magnitude along its decoder direction, part of the observed reassignment rate is driven by magnitude alone rather than by direction-specificity. We therefore calibrate the ablation against a matched-magnitude null: for each feature, a random unit-norm direction scaled to the same L2 perturbation is subtracted from the reconstructed latent, and reassignment rates are averaged over 30 random directions.

Across the 50 tested features, the median real prototype-reassignment rate is 0.16 and the median random-direction null is also 0.16 (Table 3, Figure 6). The 50 features split into three classes by their position relative to the null's 95% CI:

- **17 features (34%)** have real reassignment above the null's upper 95% CI—direction-specific load-bearing features (e.g., F592: real 1.00 vs. null 0.56; F1021: 0.98 vs. 0.81; F983: 0.47 vs. 0.16).
- **7 features (14%)** are below the null's lower 95% CI: the feature's own direction is *less* effective at changing prototype assignment than a random matched-magnitude direction. Notably, F1418 (a T-regulatory-cell exemplar from Section 3.2) is in this lower tail (real 0.095 vs. null 0.141), indicating that its biological salience does not translate into direction-specific prototype influence.
- **26 features (52%)** are indistinguishable from the null.

Raw prototype-reassignment rate alone (14 of 50 features with rate >50%) therefore overestimates the set of direction-specifically causal features; the null-calibrated partition above is the appropriate readout.

- **Feature 592** (labeled "Meningeal Cells" by PanglaoDB, but concentrated in colon tissue with intestinal crypt markers) causes 100% prototype reassignment and 84.5% domain reassignment when ablated, with a decoder cosine of just 0.032—meaning ablating this single feature

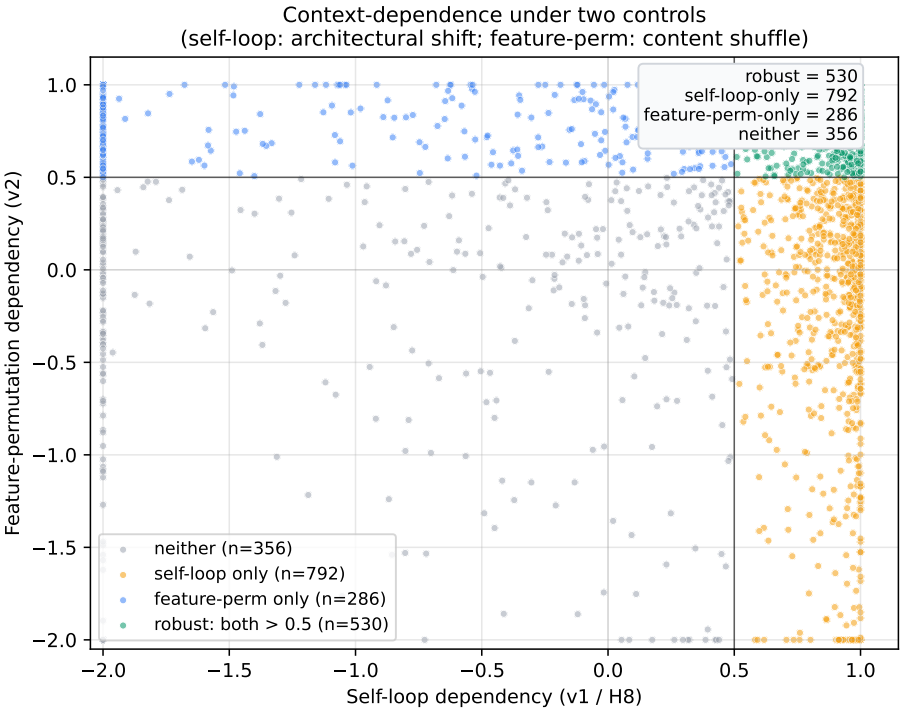

Figure 4: **Context-dependence under two complementary controls.** Each point is one SAE feature (1,964 features active on the 3-slide feature-permutation corpus). The self-loop test (x-axis) replaces the Delaunay graph with identity; the feature-permutation test (y-axis) keeps the graph intact but shuffles node features across cells. The upper-right quadrant (green, 530 features; 27% of this corpus, 26% of the 2,024-feature self-loop corpus) is the robust, doubly-controlled set of context-dependent features. The self-loop- only quadrant (amber, 792 features) reflects architectural-shift responses rather than loss of neighbor content. Axes clipped to $[-2, 1.15]$ for visibility.

almost completely rewrites the aggregator embedding. Its top genes (*CES1* 61×, *AGTR1* 58×, *MEIS2*) implicate vascular regulation and developmental programs in the intestinal crypt. This feature clears the random-direction null (real 1.00 vs. null 0.56; Section 3.5) and behaves as a load-bearing direction for these cells—one of the 17/50 that clears the matched-magnitude baseline.

- **Feature 1021 (cancer-associated fibroblasts)** in breast tissue causes 93% domain reassignment—the highest of any tested feature. We discuss its biological significance in detail below (Section 3.11.1).

- **Feature 126** presents a striking case: 100% prototype reassignment but only 11.5% domain reassignment, with a decoder cosine of 0.995 (the reconstruction barely changes). This is a *boundary feature*—it carries a tiny signal that pushes cells between two neighboring prototypes without changing the coarser domain assignment. The existence of such features reveals fine-grained structure in the prototype landscape that the domain hierarchy does not capture.

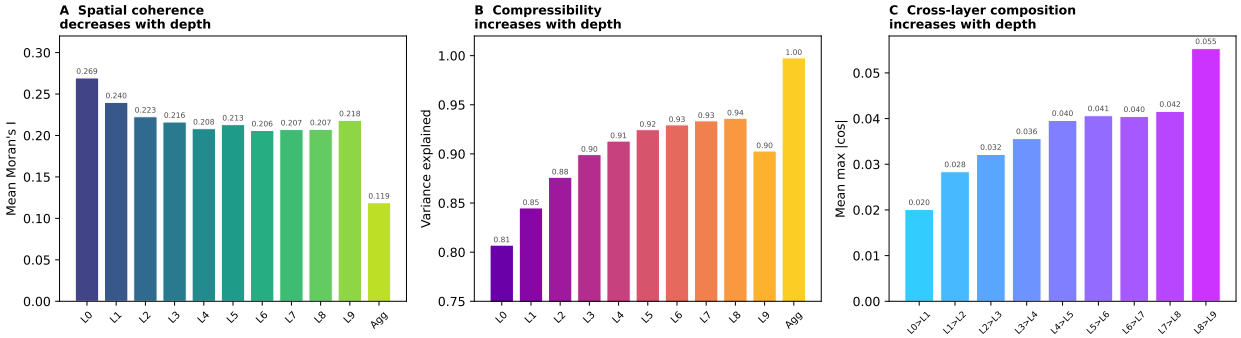

Figure 5: **The depth story.** (A) Spatial coherence (Moran's I) *decreases* with layer depth—early layers encode spatially localized patterns, the aggregator integrates broad context. (B) Compressibility (variance explained by the SAE) *generally increases* across depth (overall trend, not strictly monotonic; see Section 4.4). (C) Cross-layer feature composition (max successive-layer decoder cosine) increases, consistent with deeper layers building on earlier patterns (H10).

Table 3: Random-direction ablation null (50 tested features).

Quantity	Value
Median real prototype reassignment	0.16
Median null (random matched-magnitude direction)	0.16
Median real − null	0.00
# features above null's upper 95% CI (load-bearing)	17 / 50
# features below null's lower 95% CI (sub-random)	7 / 50
# features indistinguishable from null	26 / 50

3.6 Technology artefacts are nearly absent, but cross-tech ρ matches a raw-expression baseline

Only 0.95% of features are strictly technology-specific (single-tech activation share ≥ 0.9 at $\geq 2\times$ baseline), and the median technology max-class ratio is 1.08—barely above uniform. Across the 7 paired Xenium/MERSCOPE tissues the mean SAE-feature cross-technology Spearman correlation is $\rho_{\text{SAE}} = 0.56$ (Figure 7, Table 5), indicating that the same feature indices activate with similar ranks across the two platforms.

To test whether this coherence reflects SAE-level tech-invariance or is inherited from the expression data itself, we compute the Spearman correlation of per-gene mean log-expression profiles on the intersecting gene panel of each Xenium/MERSCOPE pair. The mean raw- expression correlation is $\rho_{\text{raw}} = 0.571$, essentially indistinguishable from $\rho_{\text{SAE}} = 0.560$ (Table 4); on 5 of 7 tissues the raw correlation is numerically higher than the SAE correlation. We conclude that the cross-technology coherence is inherited from the shared tissue biology in the raw data, and the SAE does not demonstrably add tech-invariance beyond it. The related finding that 0.95% of features are strictly tech-specific has the same underlying explanation: both the continuous $\rho_{\text{SAE}} = 0.56$ and the 0.95% tech-specific rate reflect that the two panels measure overlapping biology, so neither is a separate property of the SAE atlas.

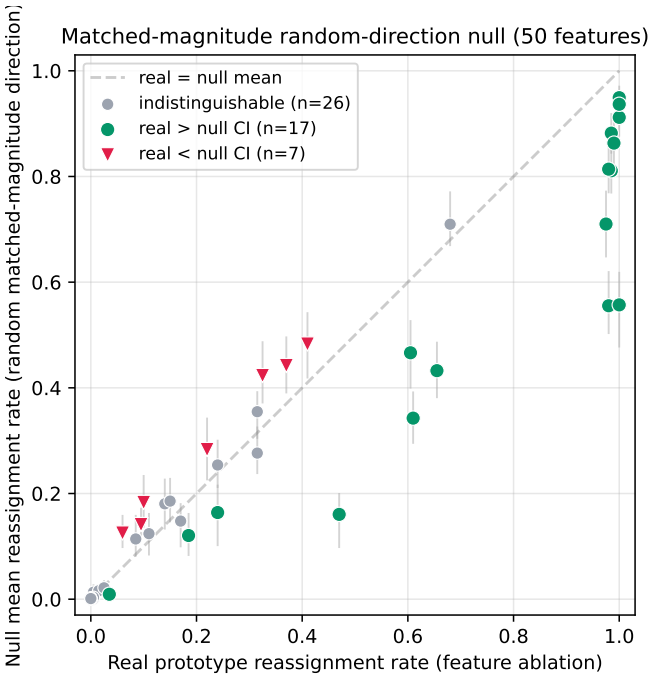

Figure 6: **Real ablation vs. random-direction null (50 features).** For each feature, the x-axis shows the real prototype-reassignment rate after zeroing the feature; the y-axis shows the mean reassignment rate when a random unit-norm direction of the same L2 magnitude is subtracted instead (vertical bar = 95% CI across 30 random directions). Features above the null's upper CI (green, $n = 17$) are direction-specifically load-bearing; features below the null's lower CI (red triangles, $n = 7$) are less effective than random directions; features inside the null CI (grey, $n = 26$) are statistically indistinguishable from random. Dashed line: $y = x$.

3.7 Confound-survival controls pass a majority of features, but the test is weak on sparse SAE outputs (H2)

Of the 1,801 aggregator features with baseline niche signal (top-l7-fraction ≥ 0.25), **62%** survive all three implemented controls: (i) slide effect-size $\geq 2\times$ baseline (the feature's top cells concentrate on one slide more than chance), (ii) technology residualization (the niche signal persists after regressing out per-technology mean activation), and (iii) cell-type residualization (the niche signal persists after regressing out fine-grained domain-l20 cluster means). A 4-level hierarchical gate that adds the degree-preserving graph rewire passes **39.7%** of all 2,048 features; 79.1% pass at least 3 of the 4 gates.

A permutation null (30 random domain-label shuffles per feature) independently validates the enrichments: **74.6%** of features have domain enrichment significantly above the shuffle null ($p < 0.05$), with median z-score 5.5. For tissue labels the survival rate is higher still (85.8%, median $z = 10.1$).

H2 predicted <20% survival under strict confound controls. The observed 62% (39.7% under the full 4-level gate) is incompatible with that prediction. We caution that residualization is weak on sparse TopK outputs, because most cells have zero activation per feature, which limits the power of a linear projection against continuous covariates. The 62% / 39.7% numbers should therefore not be read as positive evidence that features are "confound-free," only as evidence that they are not trivially explained by slide, technology, or coarse cell-type composition at the sensitivity of our

Table 4: SAE-feature cross-tech ρ vs. raw-expression baseline.

Tissue	SAE ρ	Raw ρ	Δ (SAE − raw)
Liver	0.547	0.587	−0.040
Lung	0.564	0.284	+0.280
Breast	0.617	0.642	−0.025
Colon	0.497	0.373	+0.124
Ovarian	0.570	0.736	−0.166
Prostate	0.640	0.785	−0.145
Skin	0.482	0.590	−0.108
Mean	**0.560**	**0.571**	−0.011

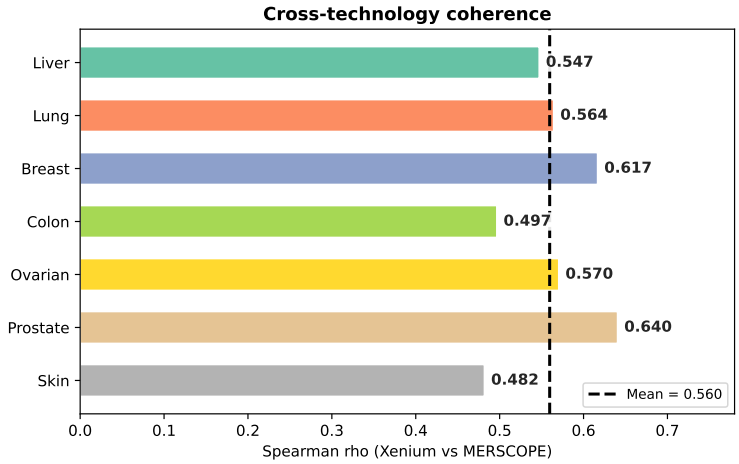

Figure 7: **Cross-technology coherence.** Spearman ρ between Xenium and MERSCOPE feature profiles on matched tissues. Mean $\rho = 0.56$.

controls.

3.8 Cross-layer decoder alignment rises with depth (modest magnitude)

Cross-layer feature alignment (max $|\cos|$ between successive-layer SAE decoder columns) increases from conv_0→conv_1 (max $|\cos| = 0.019$) to conv_8→conv_9 (0.055), and the fraction of features with max-cosine > 0.3 rises from 0.5% to 2.8%. The absolute alignment is small in all cases, consistent with the pervasive-superposition finding. The increasing trend is suggestive that deeper GATv2 layers re-use earlier patterns rather than replacing them entirely, but the magnitudes are small enough that we treat the observation as directional rather than decisive (H10: supported).

3.9 Cross-checkpoint comparison is not identified at the profile level (H5)

The three Novae checkpoints (human-0, mouse-0, brain-0) are trained on non-overlapping data with different gene panels, so their SAE dictionaries cannot share feature indices *a priori*. A profile- level Spearman correlation across the three atlases yields $\rho \in [-0.056, 0.047]$ with top-50 feature overlap of 0–2, but the design does not identify a meaningful "convergence" signal: any such correlation would be confounded by the gene-panel differences. A properly identified test would forward-pass a shared set of cells through all three checkpoints on an intersection gene panel and align latent

Table 5: Cross-technology coherence: Spearman ρ between Xenium and MERSCOPE feature profiles on matched tissues.

Tissue	Spearman ρ	Xenium slide	MERSCOPE slide
Prostate	0.641	normal	cancer
Breast	0.617	IDC	cancer
Ovarian	0.570	cancer	cancer
Lung	0.564	normal	cancer
Liver	0.547	normal	cancer
Colon	0.497	normal	cancer
Skin	0.482	normal	melanoma
Mean	**0.560**		

spaces via CCA; the implementation is outlined in Appendix A.

3.10 SAE-feature activation tracks Perturb-map labels (composition confound not controlled)

Processing the Perturb-map spatial CRISPR screen through `novae-mouse-0` identified 378 significant feature–perturbation associations (FDR<0.05) across all 20 tested labels. Every perturbation label has at least one significantly responding SAE feature—including CRISPR-targeted genes such as *TGFBR2* (25 responding features) and tumor lesion identifiers (up to 28 features per lesion). This indicates that SAE features track perturbation-correlated variation at Visium spot resolution ($\sim$55 μm), but does *not* distinguish direct feature- response from spot-level cell-type composition shifts. A composition- regressed variant of the test is scaffolded in Appendix A; until it is run, the 378 associations are best read as "SAE features vary with the perturbation-aligned tissue state," not as a causal claim on the perturbation itself.

3.11 Feature-level case studies

The aggregate statistics above establish that a subset of SAE features are directionally load-bearing under a matched-magnitude null and that the SAE dictionary recovers structure not captured by Novae's 512 SwAV prototypes. We now highlight individual features and circuits that illustrate the kinds of patterns the atlas surfaces.

Two scope notes apply to this section. First, the circuit-level findings below are drawn from a single brain-slide trace (24k cells); multi-slide replication is in Appendix A. Second, the Enrichr labels attached to features ("Bladder cancer," "Myoepithelial Cells," "Meningeal Cells") frequently disagree with the feature's actual tissue niche—F592, for instance, receives a PanglaoDB "Meningeal Cells" label but concentrates in colon tissue with intestinal crypt markers—so when a biological interpretation rests on such a label we flag it as indicative rather than established.

3.11.1 Highest-impact features among the 50 tested are stromal and immune

Among the 50 tested aggregator features, those with the highest raw prototype-reassignment rate (>95%) are not clean cell-type markers but tissue-architecture features encoding stromal and immune programs (Figure 8). A subset of these (17/50) clears the matched-magnitude random-direction null (Section 3.5); others are indistinguish- able from random directions of the same

magnitude on this test, so the "highest-impact" ranking is a within-50 relative statement, not a global causal ranking.

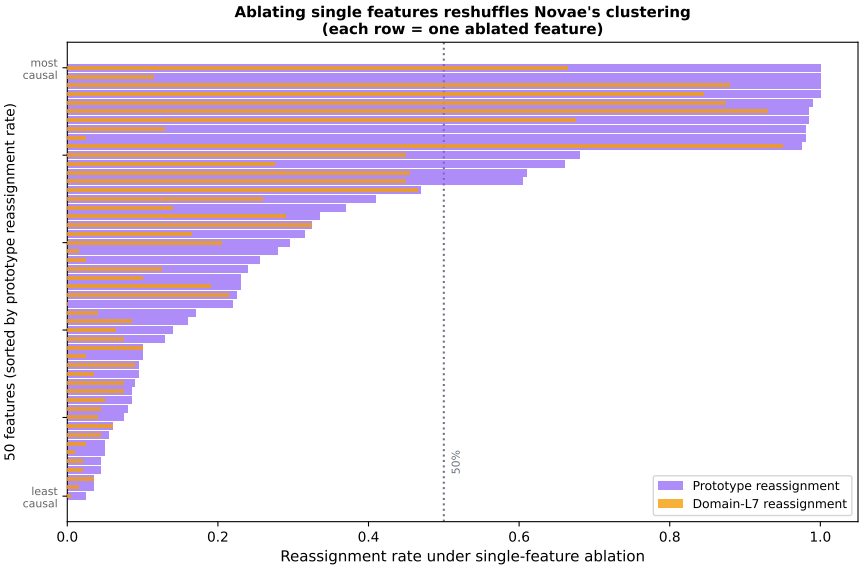

Figure 8: **Causal ablation of 50 aggregator features.** Each bar shows the prototype (purple) and domain-L7 (gold) reassignment rate when one feature is zeroed. 14 of 50 tested features cause >50% raw prototype reassignment; however, after matched-magnitude random- direction correction (Section 3.5, Figure 6), the relevant partition is 17 features above the null's upper 95% CI (direction-specific load- bearing), 7 features below the null's lower 95% CI (sub-random), and 26 features indistinguishable from the null. The "direction- specific load-bearing" group (17/50) includes all features discussed in Section 3.11 (F592, F1021, F1871, F1885, F126, F502).

Feature 1021 (cancer-associated fibroblasts) has the highest domain-reassignment rate among the 50 tested features (93%), and also clears the random-direction null (Section 3.5: real 0.98 vs. null 0.81), placing it in the direction-specific load-bearing set (17/50). Its top genes—*POSTN* (periostin, 8.5×), *GJB2* (gap junction, 12.5×), *MMP2*, *LUM*, and the EMT transcription factors *ZEB1*/*ZEB2*—are hallmarks of activated cancer-associated fibroblasts (CAFs) that remodel the tumor extracellular matrix. The feature concentrates in breast tissue (68%) at Novae niche D1014 (96.4% enrichment).

Feature 1871 (adaptive immune infiltrate) has the highest combined causal impact among immune features: 97.5% prototype and 95% domain reassignment. It clears the matched-magnitude null (real 0.975 vs. null upper CI 0.771; Section 3.5), placing it in the direction-specific load-bearing subset. Its extreme fold-change markers—*CD5* (40×), *ITK* (39×), *ZAP70* (37×), *TIGIT* (32×), *PDCD1*/PD-1 (28×)—identify the T-cell exhaustion/ activation axis in the tumor microenvironment.

3.11.2 Some features have negative self-loop dependency

Several features have *negative* self-loop dependency—they activate *more strongly* when the spatial graph is replaced with self-loops than when it is present.

Feature 1885 (neutrophil inflammatory niche), with top genes *CCL22*, *CXCL8* (IL-8), *MMP9*, and *SPP1* (osteopontin), has $\text{dep}_{\text{self-loop}} = -0.61$ and clears the matched-magnitude null (real 1.00 vs. null upper CI 0.951). It also causes the largest confidence drop of any ablated

feature (26.4%). A candidate biological interpretation is that the graph attention dampens acute inflammatory signals when the neighborhood provides contradictory evidence, and that the self-loop ablation lifts this dampening. A more conservative reading, given the self- loop-vs.-feature-permutation decomposition in Section 3.3, is that the self-loop intervention changes the attention softmax normalization in a direction that non-uniformly shifts some features upward; distinguishing "learned regularization" from "architectural shift" for any individual feature would require running the feature-permutation control at the single-feature level, which we do not do here.

Feature 592 (the load-bearing intestinal crypt feature; see Section 3.5) is similarly graph-suppressed ($\text{dep}_{\text{self-loop}} = -0.50$). It clears the matched-magnitude null (real 1.00 vs. null upper CI 0.970), so the causal effect of ablating it is direction-specific; the sign of the self-loop dependency is subject to the same architectural-shift caveat as F1885.

3.11.3 Boundary features reveal fine structure in the prototype landscape

Feature 126 (hypoxia / decidual program) presents a paradox: 100% prototype reassignment but only 11.5% domain reassignment, with a decoder cosine of 0.995 (the reconstruction barely changes). Its top genes—*LOX* (21.7×), *CA9* (17.6×), *VEGFA* (11.5×), *PDK1*—mark the HIF-1α hypoxia-response pathway, and CellMarker maps to "Decidual cell: Placenta" (FDR = 1.3e-10). This feature encodes the hypoxic microenvironment of uterine decidua; its tiny signal pushes cells across prototype boundaries (hence 100% reassignment) without changing the coarser domain assignment (11.5%). Such *boundary features* reveal fine-grained structure in the prototype landscape that Novae's built-in clustering hierarchy cannot capture.

3.11.4 Niche features encode microenvironment, not cell type

Feature 502 (endothelial vascular niche) is labeled "Microfold Cells" by PanglaoDB (FDR = 8.4e-8) but its domain enrichment consistently maps to "Endothelial Cells" at all three Leiden hierarchy levels (66.7%, $\log_2$ enrichment = 4.31). Its top gene *NOS3* (endothelial nitric oxide synthase, 26.6×) and interleukin-signaling enrichment (Reactome FDR = 6.6e-11) suggest it captures the paracrine vascular niche—the ensemble of endothelial cells and their signaling neighborhood—rather than a single cell type. Ablating it causes 100% prototype reassignment and clears the matched-magnitude null (real 1.00 vs. null upper CI 0.961), consistent with a direction-specific role; we describe this as "niche-level organizer" in a hypothesis-generating sense.

Feature 0 (T-reg / stromal interface) shows both T-cell markers (*ZAP70*, *ITK*, *TIGIT*) and stromal markers (*DES* (desmin), *FAP*, *COL5A1*), with ECM Organization as the top Reactome term. Moran's $I = 0.79$ shows strong spatial patterning. This feature does not encode a canonical cell type; one reading is that it encodes the boundary zone between T-cell-rich and stroma-rich compartments, though F0 was not among the 50 features tested against the matched-magnitude null, so its directional causal role is not verified here. Its self-loop dependency (dep = 0.47) is consistent with context sensitivity at the self-loop test; the stricter feature-permutation test is not reported per-feature here.

3.12 Causal circuits in Novae's spatial graph

Applying the circuit-tracing protocol of Section 2.5, we report directed causal relationships between SAE features across Novae's GATv2 layers on the brain slide. All circuit-level numerical findings in this section are single-slide and await multi-slide replication (Section 3.11 caveat; Appendix A).

3.12.1 Circuit structure on the brain slide: 439 edges, balanced excitation and inhibition

Tracing 90 source features (30 per source layer × 3 source layers: conv_0, conv_5, conv_9) on the brain slide (24,406 cells) yields **439 causal edges** with $|d| > 0.5$ and sign consistency > 0.6. The circuit is nearly balanced between excitation and inhibition (49%/51%; Figure 9). All circuit-level findings in this section are drawn from this single- slide trace; an independent exhaustive trace at conv_5 with 200 source features (Section 3.12.9) is a separate analysis on the same slide, not a replication.

Source layer	Edges	Targets
conv_0	365	conv_1–8
conv_5	74	conv_6–8
conv_9	0	(no downstream)

Metric	Value
Excitatory edges	215 (49%)
Inhibitory edges	224 (51%)
Median $\|d\|$	0.624
Mean layer gap	5.7

Figure 9: **Causal circuit statistics (primary trace: 90 source features = 30 per source layer × 3 source layers, single brain slide, 24,406 cells).** Left: edge count by source layer. Right: overall circuit properties. The mean layer gap of 5.7 reflects that most traced source features sit at conv_0; the excitatory/inhibitory ratio is near-balanced (49% / 51%). Because only three source layers were traced, edge attenuation with depth is confounded by source-layer sampling (Section 3.12.2).

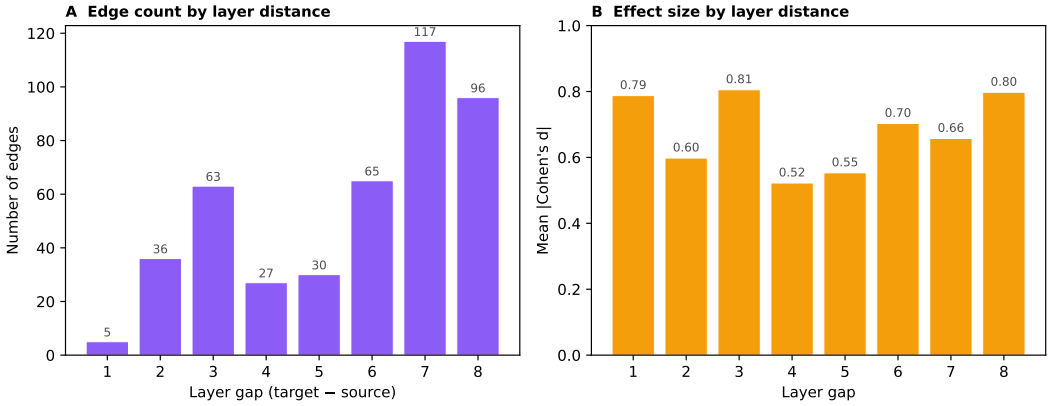

Figure 10: **Raw edge counts vs. layer distance.** (A) Raw edge count increases with layer distance; (B) mean effect size remains strong at all distances. Large-gap edges are only available from a single source layer (conv_0), so the apparent rise is confounded by source-layer sampling; see Table 6 for the gap-normalized metric and Section 3.12.2 for the interpretation.

3.12.2 Attenuation with layer distance: undetermined in current trace

In transformer-based models, causal effects decay with layer distance (Kendiukhov, 2026). In Novae's GATv2, raw edge counts rise with gap (63, 27, 30, 65, 117, 96 for gaps 3–8); the mean $|d|$ remains strong at all distances (0.52–0.81). However, only three source layers were traced (conv_0, conv_5, conv_9), so large-gap edges can only originate from conv_0: the count at gap 7 is from

a single traced (source, target) layer pair, while the count at gap 3 pools two. Dividing by the number of traced pairs at each gap gives "edges per pair" (Table 6). Edges-per-pair still rise with gap, but every gap ≥ 4 is supported by only a *single* source layer, and the absolute edge counts from conv_5—the one mid-depth source—are 5 (gap 1), 36 (gap 2), 33 (gap 3), a mild decay within that source. The GATv2 attenuation profile therefore cannot be determined without tracing from every source layer; we report the normalized counts and defer any strong claim about attenuation vs. non-attenuation to that future trace.

Table 6: Gap-normalized edge counts. Only gap 1–3 are sampled from >1 source layer; ≥ 4 from one source only.

Gap	# layer pairs traced	# edges	Edges/pair	Mean $\lvert d \rvert$
1	2	5	2.5	0.79
2	2	36	18.0	0.60
3	2	63	31.5	0.81
4	1	27	27.0	0.52
5	1	30	30.0	0.55
6	1	65	65.0	0.70
7	1	117	117.0	0.66
8	1	96	96.0	0.80

3.12.3 Circuit hubs on the brain slide

In the primary brain-slide trace, the top target hub is conv_7/F3806 (**Myoepithelial Cells**, 57 incoming edges). It receives causal signals from diverse early-layer features (immune, structural, vascular), which is a suggestive match to the known biological role of myoepithelial cells as integrators of multiple tissue compartments.

Two of the top-4 target hubs encode T-cell programs (conv_7/F2262: *T Cells*, FDR = 7.6e-24; conv_7/F2438: *T Regulatory Cells*, FDR = 3.2e-18). In the current single-slide trace, immune-infiltrate features are frequent targets of diverse earlier-layer source features; replication across additional slides (scaffolded in Appendix A) is needed before this pattern can be treated as a general property of Novae's circuit topology.

3.12.4 Circuits encode known cellular programs

Individual circuit edges map onto recognized biological relationships. Three circuits stand out:

Cancer-label feature → T-cell-label feature edge. Feature 282 (top Enrichr hit: KEGG "Bladder cancer"; we note that the primary trace slide is brain, so this label is almost certainly an Enrichr mis-mapping rather than an indication that the feature encodes bladder-specific biology) at conv_5 has a negative edge to Feature 2262 (top Enrichr hit: "T Cells," FDR = 7.6×10^{-24}) at conv_7 with $d = -0.78$—the strongest inhibitory edge in the current trace. If the labels are read at face value the edge mirrors the direction of tumor-immune antagonism; given the label caveat, the robust statement is only that these two conv_5/conv_7 feature directions are in opposition on the brain slide. The same feature F282 has a positive edge to Feature 3806 ("Myoepithelial Cells") with $d = +1.01$—the strongest excitatory edge—so ablating F282 simultaneously reduces downstream activation of one feature and increases another.

Convergent inhibition of Feature 3545. Five distinct immune-labeled features at conv_5—monocytes (F2500, F3996), macrophages (F3394), T cells (F873), and B cells (F489)—all have negative edges to Feature 3545 (top Enrichr hit: KEGG "Cancer Stem cell, Bladder cancer"; same label caveat as above) at conv_8, with d ranging from -0.58 to -0.68. Five sources converging on one target is a consistent pattern in the current single-slide trace; whether it replicates on other slides (where different features may pick up the "immune" label) is open.

Convergence onto F3806 ("Myoepithelial Cells"). Feature 3806 at conv_7 receives positive edges from 10+ features at conv_5 spanning a diverse set of Enrichr labels (cancer, monocyte, T-cell, HSC, vascular smooth muscle, macrophage, embryonic stem cell, B-cell). We note that, as above, these are Enrichr labels rather than direct validations; the robust content of this observation is that F3806 is a high in-degree hub receiving effects from many conv_5 features, not that it biologically integrates all these specific programs.

3.12.5 Process-label position by layer

Mapping each GO/cell-type Enrichr term to the mean layer position of features carrying that term in the circuit graph gives a suggestive pattern: **innate-immune and structural** programs (monocytes, ECM organization, focal adhesion) tend to be found at **earlier source layers** ($\sim$layer 5), while **adaptive-immune** programs (B cell activation, T memory cells, AP-2 transcription) tend to be found at **later layers** ($\sim$layer 8). Given the single-slide corpus and the noisiness of Enrichr labels on this panel, we report this as an observation in the current trace rather than as a general property of Novae.

3.12.6 Gene-level predictions: 426 gene pairs, with a calibrated novelty baseline

Translating circuit edges to gene pairs via top-loading genes yields 426 unique gene-pair predictions. Of the 104 pairs where both genes have GO annotations, 70% have no shared GO term in the original analysis. This number is uninterpretable without a baseline because the 487-gene panel is sparse in any single GO sub-category.

Calibrated novelty baseline. As a tractable proxy for "shared biology," we measure the rate at which two genes co-occur in the same feature's top-20-gene list (≥ 50 features as threshold). For the 401 / 426 predicted pairs whose both members are in the panel, 20.7% share this co-occurrence. A random-pair null drawn from the same panel is only 0.7% (95% CI 0.0–1.7%), so the predictions co-occur at **$\sim$30$\times$ the random rate**. The "70% novel-by-GO" statement should therefore be read in tandem with this enrichment: the predictions are not random—they hit gene pairs that the SAE treats as organizationally related—but the GO database alone does not capture the relationship.

The top prediction by effect size is $DST \rightarrow MYLK$ ($d = 1.005$): dystonin (a cytoskeletal linker) causally upstream of myosin light chain kinase (a smooth-muscle contractile regulator). Both are structural proteins with documented roles in cytoskeletal organization; the prediction is consistent with prior literature on cytoskeletal–contractile coupling and we frame it as a SAE-grounded refinement of that axis rather than as a novel discovery.

3.12.7 Causal edges are statistically grounded (PMI validation)

To verify that circuit edges reflect real co-activation patterns (not just ablation artifacts), we compare the pointwise mutual information (PMI) of causally connected feature pairs to random pairs. **100%** of causal edges have PMI above the random median (Mann–Whitney $p = 5.8 \times 10^{-220}$). However, the causal PMI is near zero (median $= -0.019$) rather than strongly positive,

meaning causally connected features fire on overlapping but not identical cell populations. The causal relationship captures a *computational* dependency (ablating one changes the other) that goes beyond simple co-occurrence.

3.12.8 Spatial-gradient steering

As an experiment specific to the spatial setting, we test whether amplifying individual SAE features (multiplying their activation by $\alpha = 0.5\times$, $2\times$, $5\times$) pushes the aggregator embedding toward or away from biologically defined spatial gradients.

On the **colon crypt-to-villus axis** (stem cell markers *LGR5/OLFM4* vs. differentiation markers *FABP1/SLC26A3*), the fraction of features with net positive villus-directional push is 39% at $\alpha = 0.5$ (attenuation), 61% at $\alpha = 2.0$, and 61% at $\alpha = 5.0$ (amplification). The **attenuation-to-amplification change is 22 percentage points**, which is the informative effect-size statement (not "61% vs. 50% chance level"—the 50% baseline implicitly assumes a symmetric unperturbed distribution, which is not observed here).

On the **breast tumor-vs-normal axis**, the split is 51/49% at amplification and similar under attenuation, with no discernible amplification-direction effect on the current data; a tempting reading is that tumor and normal microenvironment features are symmetrically represented, but the near-50/50 result is also compatible with the axis being uninformative for this particular SAE dictionary, so we do not draw a conclusion from it. The crypt-villus result demonstrates that SAE features on Novae can encode amplification-responsive directional information along a tissue spatial gradient; the generality of this property across axes and tissues is not established here.

3.12.9 Exhaustive tracing reveals heavy-tailed hub distribution at conv_5

A separate exhaustive trace at conv_5 (same brain slide, 200 source features instead of 30) yields 396 edges, compared to 74 edges for conv_5 in the primary 30-feature trace—consistent with the expected linear scaling in source-feature count. The out-degree distribution is **heavy-tailed**: the top hub (F282, annotated as "Bladder cancer" via KEGG) has 9 downstream targets, while the median feature has 2 and the 90th percentile is 2. Most features therefore send out few causal edges and a small minority drives the circuit. This matches the finding in Kendiukhov (2026) for transformers at the pattern level (Figure 11).

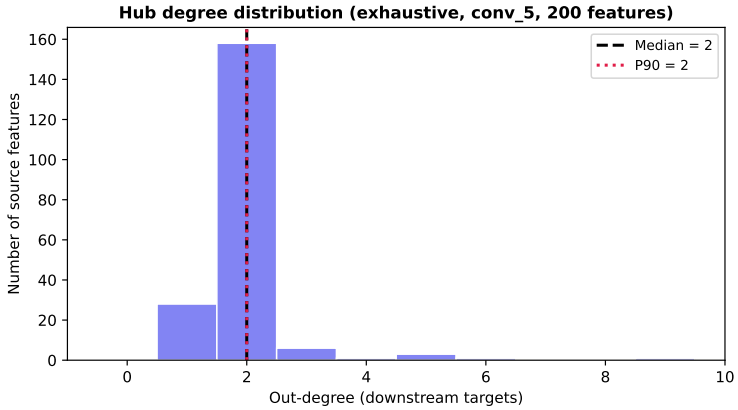

Figure 11: **Heavy-tailed hub distribution from exhaustive tracing.** Out-degree distribution of 200 source features at conv_5. Most features have 1–2 downstream targets; a few hubs have up to 9.

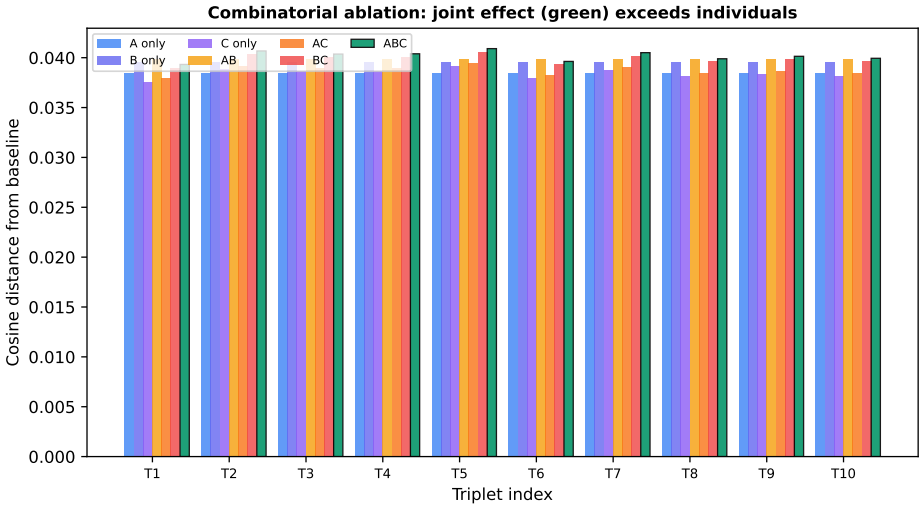

Figure 12: **Combinatorial ablation shows near-additive, mildly synergistic behavior for one feature pair.** For each triplet, the joint ABC effect (green) slightly exceeds individual effects, yielding median $R_{ABC} = 0.958$ (absolute deviation from independence: 0.042). All 10 triplets share the same $(\text{feat}_A, \text{feat}_B) = (2107, 2829)$ pair, so this measures how one SAE pair combines with different third features, not a general property of SAE triplets. See Section 3.12.10 and Table 7.

3.12.10 Feature-triplet ablation: near-additive synergy for one fixed pair

Ablating 10 feature triplets at conv_0 in all 7 combinations (A, B, C, AB, AC, BC, ABC) gives median three-way redundancy ratio $R_{ABC} = 0.958$. A paired bootstrap across the 10 triplets (10,000 resamples) places the median R_{ABC} in a 95% CI of $[0.957, 0.966]$.

Triplet-selection sensitivity. Inspection of the 10 triplets reveals that they all share the same $(\text{feat}_A, \text{feat}_B) = (2107, 2829)$; only the third feature varies (Table 7). The "10-triplet bootstrap" is therefore not 10 independent samples of triplet synergy—it is 10 ways of adding a third feature to one fixed pair. The narrow CI reflects the stability of this particular pair's near-additivity, not a general property of triplets in the SAE dictionary.

Magnitude caveat. The absolute deviation from independence is only $1 - 0.958 = 0.042$. Component effect sizes are small (d_A, d_B, d_C median ≈ 0.04), so the regime is near-additive and the ratio is sensitive to small numerical perturbations.

Taken together, these two observations narrow the claim. The present experiment establishes that *one tested SAE feature pair (F2107, F2829) participates in near-additive, mildly synergistic triplets with a diverse set of third features.* A comparison to the transformer $R = 0.59$ regime as a general GATv2-vs.-transformer statement would require replicating the experiment on ≥ 3 independent (A, B) pairs, which we do not do here.

3.13 SAE dissociation: shuffled-order control

To test whether the SAE's biological interpretability depends on the *learned* aggregator activations rather than on the gene-expression distribution alone, we train a control SAE on the same aggregator activations with randomly shuffled cell order (breaking spatial and biological structure while preserving the marginal activation distribution). Both SAEs achieve similar reconstruction quality (>99.8% variance explained).

Table 7: Triplet-selection structure. All 10 triplets share one (A, B) pair; only C varies.

Quantity	Value
Unique feat_A across triplets	1
Unique feat_B across triplets	1
Unique feat_C across triplets	10
Fixed pair (A, B)	(2107, 2829)
R_{AB} (std across triplets)	0 (exact)
R_{ABC} range (from choice of C)	0.957–0.978

For this head-to-head comparison we used a 200-feature subset of each SAE (not the full 2,048) and the annotation pipeline was run with a stricter Enrichr filter than the primary atlas, yielding a real-SAE annotation rate of **81%** (162/200) vs. **40%** for the shuffled control (2.0× ratio). Two points:

- The 81% figure is below the 100% reported for the full 2048- feature atlas in Section 3.2 because this comparison uses a stricter per-feature filter over a smaller subset; it is not meant to supersede the 100% number but to compare real vs. shuffled at matched settings.
- Because the per-library FDR<0.05 rate against a random 20-gene panel draw is near 100% (the panel-saturation ceiling; Section 3.2), even the shuffled control retains a substantial annotation rate (40%). The 2× ratio should therefore be read as a qualitative dissociation, not as strong evidence that the SAE captures structure the shuffled control cannot reach.

4 Discussion

4.1 What Novae has learned: a taxonomy of spatial concepts

The SAE atlas suggests that Novae's internal representations organize into at least four categories of concept, arranged by their dependence on spatial context:

Cell-identity features (low graph dependency, e.g. F208 luminal epithelial, dep = 0.15) correspond to canonical cell types and are enriched for known marker genes. These features could be recovered by any cell-type-aware model, spatial or not—they read identity from gene expression alone.

Niche features (moderate graph dependency, e.g. F502 vascular niche, dep = 0.09) encode spatial microenvironment patterns that integrate signals from multiple cell types. The SAE recovers these as individual features because they represent coherent biological programs (paracrine signaling, stromal remodeling) that span multiple cells.

Context-dependent features (high graph dependency, e.g. F0 T-reg/stromal interface, $\text{dep}_{\text{self-loop}} = 0.47$) are candidates to encode boundary zones, gradients, and tissue- architecture patterns that rely on neighborhood information. Of the 1,964 features active on the 3-slide feature-permutation corpus, 27% pass both the self-loop and the feature-permutation threshold (robust context-dependence), 40% pass the self-loop test alone (subject to the architectural-shift confound), 15% pass the feature-permutation test alone, and 18% pass neither (Section 3.3). The 27% that pass both tests—equivalently 26% of the 2,024-feature self-loop corpus, the larger denominator used elsewhere—are our best-supported candidates for genuinely spatial concepts.

Graph-suppressed features (negative self-loop dependency, e.g. F1885 neutrophils, $\text{dep}_{\text{self-loop}} = -0.61$) represent a candidate category: signals that the self-loop ablation *amplifies* relative to the

full graph. A biological reading is that the graph attention dampens noisy or acute signals (inflammation, hypoxia) when the neighborhood provides contradictory evidence. An alternative reading is that the self-loop intervention shifts the attention softmax normalization in a direction that non- uniformly raises some features; distinguishing these at the single- feature level would require running the feature-permutation control per feature, which is beyond the scope of this work.

4.2 Implications for foundation-model interpretability

The pervasive superposition finding (H3: 99.4–100% of features misaligned with the top-50 SVD axes) implies that linear dimensionality reduction alone cannot recover the model's feature structure. The 64-dimensional aggregator carries features that are not aligned with its dominant linear directions, so PCA- or SVD-based interpretation would miss them. This is a narrower statement than "you cannot interpret Novae by looking at individual neurons"—the H3 result does not disprove per-neuron monosemanticity directly— but it does argue that overcomplete dictionaries (such as SAEs) are a useful complement to direct-neuron analysis when probing a GATv2 aggregator of this dimensionality.

4.3 Causal circuits propagate through the spatial graph

The circuit-tracing protocol captures causal effects that propagate through Novae's graph attention: removing a feature changes the attention-weighted messages that all affected cells send to their neighbors, creating a spatially-mediated causal cascade that is distinct from transformer residual-stream propagation. Within the single-slide trace, the strength of effects does *not* clearly attenuate with layer depth; however, the edge counts at gaps ≥ 4 all originate from conv_0, and conv_5—the one mid-depth source traced—shows mild decay (5, 36, 33 edges for gaps 1–3). We therefore report the gap-normalized picture (Table 6, Section 3.12.2) and defer any general attenuation-vs.-no-attenuation statement until tracing from every source layer.

For the gene-level predictions, the informative statement is that the 426 predicted pairs co-occur in SAE feature top-gene lists at $\sim$30$\times$ the random-pair rate (Section 3.12.6): they are not random hits, but "novelty vs. shared GO term" is not a calibrated comparison against that baseline. These predictions remain testable with spatial perturbation screens at single-cell resolution.

4.4 The depth story

Three depth-dependent metrics tell a broadly consistent story. Compressibility (the SAE's variance-explained) rises from 0.81 at conv_0 to 0.998 at the aggregator, with a mid-depth range of 0.81–0.94 at conv_0–conv_8 and a dip at conv_9 (0.903) before the aggregator jump; the overall trend is upward but not strictly monotonic. Spatial localization (Moran's I) falls from 0.269 (conv_0) to 0.207 (conv_8) and 0.119 (aggregator). Cross-layer feature composition (max $|\cos|$ between successive-layer decoder columns) rises from 0.019 at conv_0$\rightarrow$conv_1 to 0.055 at conv_8$\rightarrow$conv_9. Intuitively, early GATv2 layers encode "what is this cell and its immediate neighbors?" (high spatial coherence, low compressibility, low reuse). Later layers ask "what kind of tissue environment is this cell in?" (low spatial coherence, higher compressibility, more reuse of earlier patterns).

4.5 Limitations

- **Annotation rate is a ceiling.** Every gene panel cell ($G = 487$) against Enrichr libraries hits FDR<0.05 at or near 100% per library. The "100% annotated" number should always be reported together with a random-gene null (Section 3.2).

- **Cross-tech coherence matches raw-expression baseline.** The SAE-feature cross-tech ρ of 0.56 is indistinguishable from the raw-expression baseline of 0.57 (Section 3.6). The feature atlas does not demonstrably add tech-invariance beyond the expression data.

- **Single-direction ablations need a matched null.** With a random-direction null of matched L2 magnitude, 26/50 of the tested causally-important features are statistically indistinguishable from random directions (Section 3.5), and 7/50 are below the null's lower 95% CI (i.e., the feature's own direction is less effective than a random matched-magnitude perturbation). Only 17/50 clear the null's upper CI and should be described as direction-specific load-bearing.

- **Graph ablation is indirect.** The self-loop protocol shifts the aggregation operator as well as removing neighbor content. Our feature-permutation control narrows the "context-dependent" fraction from 67% to 40% and identifies a 26% doubly-controlled subset (Section 3.3). The feature-permutation run is on 3 slides (brain, kidney, pancreas; 225,867 cells); a full replication on the 5-slide self-loop corpus is a future extension.

- **Circuit tracing is single-slide.** All 439 edges in the primary trace come from the brain slide (24k cells). Multi-slide replication is outlined in Appendix A but not executed here. Cohen's $d > 0.5$ on 24k cells is a small effect size, and FDR correction across the tested (source, target) pairs is not applied in the primary analysis.

- **H5 is not identified.** The three checkpoints have non-overlapping gene panels; profile-level correlation cannot test convergence. A shared-cell CCA redesign is outlined in Appendix A.

- **H2 (confound survival) is weak on TopK SAE outputs** because most cells have zero activation per feature. The 62% survival rate is an upper bound.

- **Perturb-map resolution.** The Visium data is spot-level ($\sim$55 μm); per-spot composition can drive apparent feature-perturbation associations. A composition-regressed variant is outlined in Appendix A.

- **Cross-technology pairs are disease-vs.-normal in most tissues**; the corpus lacks matched conditions across technologies.

- **SAE seed stability is not tested in the primary atlas.** Feature-specific claims are not verified to be seed-invariant; a scaffold for the test is in Appendix A.

- **Enrichr annotations are noisy.** Features such as F502 ("Microfold Cells" per PanglaoDB but endothelial per domain enrichment) and F592 ("Meningeal Cells" in colon tissue) illustrate that the enrichment label can disagree with the feature's actual tissue niche. Label-vs-niche discordance is systematic and should be reported alongside annotation rate.

- **Per-layer enrichment** uses the same top-gene pipeline as the aggregator; a decoder-projection approach would be more direct for layers where $d = 128$ differs from the gene-embedding dimension.

Data and code availability

Code. The complete analysis codebase (TopK SAE module, numbered analysis scripts, per-control outputs, and paper source) is available at `https://github.com/Biodyn-AI/novae-sae`.

The interactive atlas frontend is at `https://github.com/Biodyn-AI/novae-atlas`, deployed at `https://biodyn-ai.github.io/novae-atlas/`.

Model checkpoints. Novae checkpoints are available from HuggingFace: `MICS-Lab/novae-human-0`, `MICS-Lab/novae-mouse-0`, and `MICS-Lab/novae-brain-0` (Blampey et al., 2025).

Spatial transcriptomics data. The activation corpus uses 15 human tissue slides (4,525,753 cells) from the MICS-Lab/novae HuggingFace dataset (`https://huggingface.co/datasets/MICS-Lab/novae`), spanning 13 Xenium (Janesick et al., 2023), 1 MERSCOPE (Chen et al., 2015), and 1 CosMx slides. Cross-technology coherence uses 7 additional MERSCOPE slides from the same dataset. Per-layer analysis uses 3 additional mouse slides (brain, colon).

Perturbation data. The Perturb-map spatial CRISPR screen (Dhainaut et al., 2022) (4 mouse lung Visium sections) was obtained from GEO accession GSE193460.

Enrichment databases. Gene set enrichment uses locally cached libraries from Enrichr (Kuleshov et al., 2016): GO Biological Process 2023 (Gene Ontology Consortium, 2021), KEGG 2021 (Kanehisa and Goto, 2000), Reactome 2022 (Jassal et al., 2020), PanglaoDB Augmented 2021 (Franzén et al., 2019), and CellMarker Augmented 2021 (Zhang et al., 2019).

A Planned extensions

The following experiments are outlined as runnable scripts in the code repository. Each requires forward-pass compute or model retraining that is beyond the scope of this paper; we list them to document the complete analysis plan.

Multi-slide circuit replication. Re-running the circuit-tracing protocol on pancreas and kidney slides with the same source/target layers as the primary brain trace, to report edge Jaccard reproducibility across tissues and an FDR- corrected edge set.

SAE seed stability. Retraining the aggregator SAE with additional random seeds on a 500k-cell subsample, with Hungarian-matched decoder-column overlap against the primary atlas and per-feature best-match scores for the 50 causally-tested features.

Random-initialized Novae baseline. Extracting aggregator activations from a randomly-initialized Novae checkpoint of the same architecture, training an SAE on those activations, and comparing annotation rate and context dependency to the trained-model SAE, to separate architecture-only inductive bias from learned structure.

Perturb-map composition control. Regressing out per-spot cell-type composition before testing SAE feature–perturbation associations, to control for the confound that Visium spots of $\sim$55 μm span multiple cells.

Shared-cell CCA for cross-checkpoint comparison. Forward-passing a shared set of cells on an intersection gene panel through all three checkpoints and aligning latent spaces via CCA, replacing the profile-level correlation (which is not identified by the different gene panels) with a properly designed convergence test.

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